

Proposal: Web-Based Dashboard for Sensor Testing and Analysis

I propose developing a simple web-based dashboard to visualize and monitor the status and performance of sensors during different stages of testing and assembly. This will complement the existing database by providing clear and interactive visual summaries.

1. Sensor Availability Tracking

We will implement a histogram (stacked bar chart) to track sensor flow through different stages of the workflow. Each stage will be represented as a bin, such as:

- DCA
- NISER Received
- Tail Bonding
- HV/IV Testing
- Module Production

For each stage:

- The **total available sensors** will be shown in one color
- The **number of sensors passed to the next stage** will be shown in another color

This will help identify bottlenecks and monitor overall progress.

2. Leakage Current Distribution (Overall)

A histogram will be used to show the distribution of leakage current across all sensors. Suggested bins:

- 0-1000
- 1000-2000
- 2000-10000
- 10000

This provides a quick overview of sensor quality and helps flag abnormal behavior.

3. Leakage Current Distribution (Per Batch)

Sensors are received in batches (typically ~80 sensors per batch). In addition to the overall distribution, we will include:

- A **per-batch leakage current histogram**
- Ability to **select and view individual batches**
- Comparison between **batch-level and overall distributions**

This will help identify batch-specific issues and variations in sensor quality.

4. Multi-Test Comparison per Sensor

Each sensor is tested multiple times:

- VQC (Vendor Quality Check)
- SQC (Site Quality Check)
- NISER Measurement

We will create a combined plot where:

- Each sensor is identified by its sensor ID
- Measurements from different stages are displayed together

This allows direct comparison and helps study consistency and systematic differences between measurements.

Objective

The dashboard will:

- Provide a clear overview of sensor workflow and status
- Help monitor quality through leakage current distributions
- Enable batch-level diagnostics
- Allow comparison of measurements across different testing stages

A demo version will be developed first using sample data to validate the design, followed by integration with the existing database.
